

Thirupathi R. Vajjala

Principal Engineer | Applied AI/LLM Engineering | Cloud-Native Distributed Systems

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Executive profile

Principal Software Engineer with 18+ years of experience building, modernizing, and operating large-scale distributed systems across Oracle Cloud Infrastructure, financial services, and enterprise platforms. **Hands-on applied AI/LLM engineer** with experience building custom MCP servers, RAG and LangChain pipelines, prompt-engineered operational workflows, and private/local LLM development environments. Combines architecture leadership, forward deployment, production delivery, customer onboarding, and mentoring with deep expertise in cloud-native services, APIs, automation, observability, and test-driven engineering.

AI / ML & engineering capabilities

- Generative AI & LLM Engineering: prompt engineering, retrieval-augmented generation (RAG), LLM agents, local/private model workflows (Ollama), agentic coding with Codex.
- Agent Integration & Orchestration: custom Model Context Protocol (MCP) servers, LangChain pipelines, LangGraph agent-orchestration concepts, tool calling, contextual retrieval, and operational intelligence use cases.
- Forward Deployment & Customer Integration: translate operational needs into deployable solutions; automate customer onboarding with Terraform providers; integrate CI/CD, Kubernetes, telemetry, reliability, and incident-response workflows into production environments.
- Distributed Systems at Scale: multi-tenant cloud services, microservices, asynchronous/event-driven architectures, Redis/PostgreSQL/Cassandra, API platforms, performance tuning, and automated testing.

Professional experience

Oracle | Principal Engineer | Austin, TX

Mar 2020 - Present

- Lead engineer for OCI Redis and PostgreSQL managed cloud services; drive design, scalability, reliability, and operations for multi-tenant service infrastructure. Previously supported Oracle Fusion Apps Storage / Block Volume Lifecycle Management.
- Built Python custom MCP servers that enable natural-language querying of OCI alarms, subscribed regions, and cloud resources through LLM agents; developed LangChain-based RAG workflows for operational intelligence over OCI telemetry.
- Established a private AI development stack using Ollama, Cline, and custom MCP components for air-gapped or no-external-API environments; applied Codex-based agentic workflows to code generation and refactoring.
- Performed forward-deployment work by translating customer and operational requirements into production workflows using Java, Python, Go, Dropwizard, Terraform, Jenkins, OCI SDK, and Kubernetes; created a Go-based Terraform provider to automate customer onboarding and cloud resource provisioning.
- Improved delivery confidence through test-driven development with JUnit5 and WireMock, integration-test automation, plugin-based validation, and CI/CD pipelines; partnered with cross-functional engineering and stakeholder teams.

Barclaycard US | AVP, Application Development Lead | Whippany, NJ

Jan 2019 - Mar 2020

- Led the Chargeback and Fraud Detection modules and modernization of monolithic applications into cloud-native microservices.
- Automated chargeback and fraud workflows through VISA/Mastercard API integrations, reducing manual processing by 60%.
- Introduced a Drools decision-table rule engine, RxJava reactive services, Kafka event streams, and layered test strategies across CI/CD pipelines.

Virtusa / PNC Bank | Lead Consultant | Pittsburgh, PA

Aug 2017 - Jan 2019

- Designed and delivered Open Banking API microservices on Kubernetes, integrating REST services with IIB SOAP backends and IBM API Connect gateway transformations.
- Applied CQRS, Saga, and Aggregator patterns; implemented consumer-driven contract testing with WireMock, TDD/pair programming with Spock, and Gatling performance tests.
- Worked across engineering and business partners to define foundational microservice requirements, deployment scripts, security, and delivery standards.

Innominds / Lyris | Principal Engineer | Hyderabad, India

Apr 2014 - Aug 2017

- Drove architecture and delivery of distributed digital-marketing platform services using Mule ESB, Spring Boot, ActiveMQ, Hazelcast, OAuth2, and RAML-defined REST APIs.
- Built event-based services, centralized ELK observability, REST security controls, and high-throughput caching patterns for customer engagement workloads.

Adaequare / Tria Beauty | Associate Architect | Hyderabad, India

Jan 2010 - Mar 2014

- Designed and delivered global e-commerce capabilities across customer, order, product, payments, and legacy-system integrations using Java, Hybris, Spring, iBatis, SQL Server, and MQ.
- Led requirements analysis, architecture patterns, implementation, production support, and developer guidance for Korea and US online-store channels.

Mann India Technologies / Panafoto | Software Engineer | Hyderabad, India

Aug 2008 - Dec 2009

- Developed retail sales and inventory-management modules for a nine-outlet electronics chain, including role-based security, barcode generation, distributed services, and UML documentation.

Technical skills

AI / LLM: Prompt Engineering, RAG, LangChain, LangGraph, MCP Servers, LLM Agents, Ollama, Cline, Codex

Languages: Java, Python, Go, Shell

Cloud / Platform: Oracle Cloud Infrastructure (OCI), AWS, Kubernetes, Docker, Helm, Terraform, Jenkins, Linux

Architecture: Distributed Systems, Microservices, REST, gRPC, GraphQL, SOAP, Event-Driven Systems, Kafka, ActiveMQ, API Design

Data: Redis, PostgreSQL, Cassandra, MySQL, MongoDB, Hazelcast

Quality / Delivery: TDD, JUnit5, WireMock, Spock, RestAssured, Contract Testing, Gatling, JMeter, CI/CD, Performance Tuning

Education & certifications

Bachelor of Engineering, Electronics & Communication Engineering | JNTU, Hyderabad | 2002 - 2006

| Oracle Cloud Infrastructure 2023 Certified Security Professional | OCI 2023 Architect Associate | OCI 2023 Developer Professional